Theoretical Assessment

Computer Networks I – Ordinary Call 2024/25

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Theory Group A

A1. Starting network address and broadcast

Base IP assigned: 67.54.138.55/21

To get the network IP, we must take the mask (21) number of bits of the original IP, expressed as binary, and then fill the rest out with 0, as in:

67.54.138.55 = 01000011.00110110.10001010.00110111

Original IP	01000011.00110110.10001010.00110111
Mask bits (21 ones)	**11111111.11111111.11111**000.00000000
Resulting network IP	**01000011.00110110.10001**000.00000000
Resulting IP in base 10	67.54.136.0/21

Effectively making an AND operation between those two.

And for the broadcast, you fill the last bits outside the mask with ones,

In this case the 32 – 21 = 11, so the last 11 bits.

Resulting network IP	**01000011.00110110.10001**000.00000000
Broadcast	00000000.00000000.00000111.11111111
Resulting broadcast IP	**01000011.00110110.10001**111.11111111
Resulting IP in base 10	67.54.143.255/21

A2. Network architecture & A3. Addressing

As the mask of the original network is 21 bits, there are 11 bits left for subdividing, and because of the required sizes, the first bit will decide whether the IP is of the first network S1, having 10 bits left and that way having a size of 1024 possible Ips (only 1022 usable because of network IP and broadcast IP)

Original network IP	67.54.136.0/21
	01000011.00110110.10001000.00000000
S1 network IP	67.54.136.0/22
	01000011.00110110.10001000.00000000
S1 broadcast IP	67.54.139.255/22
	01000011.00110110.10001011.11111111
S1 range of assignable IP addresses	67.54.136.1/22 ... 67.54.139.254/22

If the first subnet bit is a 1, then the second determines if the IP is of the second network S2, this time having 9 remaining bits, so 512 IPs (510 usable ones).

Original network IP	67.54.136.0/21
	01000011.00110110.10001000.00000000
S2 network IP	67.54.140.0/23
	01000011.00110110.10001100.00000000
S2 broadcast IP	67.54.141.255/23
	01000011.00110110.10001101.11111111
S2 range of assignable IP addresses	67.54.140.1/23 ... 67.54.141.254/23

Lastly, I subdivide the remaining IPs once more because of the assessment size specifications, and the last range is 8 bits, giving 256 Ips (254 usable)

Original network IP	67.54.136.0/21
	01000011.00110110.10001000.00000000
S3 network IP	67.54.142.0/24
	01000011.00110110.10001110.00000000
S3 broadcast IP	67.54.142.255/24
	01000011.00110110.10001110.11111111
S3 range of assignable IP addresses	67.54.142.1/24 ... 67.54.142.254/24

And for in-between router networks, the R3-R4 network still has to be derived from the original one, so I take the first 4 of the last available spaces:

Original network IP	67.54.136.0/21
	01000011.00110110.10001000.00000000
R3-R4 network IP	67.54.143.0/30
	01000011.00110110.10001111.00000000
R3-R4 broadcast IP	67.54.143.3/30
	01000011.00110110.10001111.00000011
R3-R4 range of assignable IP addresses	67.54.143.1/30 ... 67.54.143.2/30

The R1-R2 and R2-R3 are private ones, so I chose 192.168.0.0/30 and 192.168.0.4/30 respectively.

Original network IP	67.54.136.0/21
	01000011.00110110.10001000.00000000
R1-R2 network IP	10.0.0.0/30
	00001010.00000000.00000000.00000000
R1-R2 broadcast IP	10.0.0.3/30
	00001010.00000000.00000000.00000011

R1-R2 usable IP range	10.0.0.1/30 10.0.0.2/30
R2-R3 network IP	10.0.0.4/30
	00001010.00000000.00000000.00000100
R2-R3 broadcast IP	10.0.0.7/30
	00001010.00000000.00000000.00000111
R2-R3 usable IP range	10.0.0.5/30 10.0.0.6/30

A4. IP address assignment

For S1, R1 is assigned to IP 67.54.136.1/22

For R1-R2, R1 is assigned to IP 10.0.0.1/30

For R1-R2, R2 is assigned to IP 10.0.0.2/30

For S2, R2 is assigned to IP 67.54.140.1/23

For R2-R3, R2 is assigned to IP 10.0.0.5/30

For R2-R3, R3 is assigned to IP 10.0.0.6/30

For S3, R3 is assigned to IP 67.54.142.1/24

For R3-R4, R3 is assigned to IP 67.54.143.1/30

For R3-R4, R4 is assigned to IP 67.54.143.2/30

A5. Routing tables

R1 routing table

Mask	Destination	Next hop	Interface
255.255.252.0	67.54.136.0	Direct delivery	eth0
255.255.255.252	10.0.0.4	Direct delivery	ppp0
0.0.0.0	0.0.0.0	10.0.0.2	ppp0

R2 routing table

Mask	Destination	Next hop	Interface
255.255.255.0	67.54.140.0	Direct delivery	eth0
255.255.255.252	10.0.0.0	Direct delivery	ppp0
255.255.255.252	10.0.0.4	Direct delivery	ppp1
0.0.0.0	0.0.0.0	10.0.0.6	ppp1

R3 routing table

Mask	Destination	Next hop	Interface
255.255.255.0	67.54.142.0	Direct delivery	wlan0
255.255.255.252	10.0.0.4	Direct delivery	ppp0
255.255.255.252	67.54.143.0	Direct delivery	ppp1
0.0.0.0	0.0.0.0	67.54.143.2	ppp1

R4 routing table

Mask	Destination	Next hop	Interface
255.255.255.252	67.54.143.0	Direct delivery	ppp0
255.255.255.252	10.0.0.4	Direct delivery	ppp0
0.0.0.0	0.0.0.0	Internet	-